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Automated extraction of interfacial dislocations and disconnections from atomistic data

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ABSTRACT

We introduce Interfacial Line Defect Analysis (ILDA), a method for identifying and extracting interfacial dislocations and disconnections with little-to-no input from the user on the nature of the interface. By simply providing the orientations and coherency strains for the crystals in a coherent reference state of the interface, ILDA provides exact Burgers vectors. Alternatively, these orientations and strains can be estimated using local atomic deformation gradients, making ILDA fully automated and providing estimated Burgers vectors. ILDA also determines the step height associated with each defect line segment, in case the associated defect is a disconnection. The heart of the method is the identification of atoms residing at co-incidence sites between the two crystals and the construction of a surface mesh connecting these sites that is used to compose Burgers circuits and insert defect line segments. We demonstrate the performance of ILDA in two test cases: a twist grain boundary and a phase boundary.

1. Introduction

Line defects such as dislocations and disconnections are a ubiquitous feature of crystalline interfaces [1]. In grain boundaries (GBs), line defects may be present to accommodate misorientations from co-incidence site lattice orientations (so-called Σ boundaries). Grain boundary line defects may also form when lattice dislocations impinge on GBs, leading to dislocation transmission and/or absorption events that generate grain boundary dislocations and/or disconnections. These defects are known to affect many phenomena in materials science, including GB strengthening (e.g., Hall–Petch strengthening) [2], GB sliding [3], GB diffusion [4], intergranular fracture [5], and GB phase/complexion transitions [6]. In phase boundaries, line defects may form for similar reasons. However, unlike GBs, the two crystals forming a coherent phase boundary do not mate-up perfectly; the two lattices must be distorted to form the coherent interface. This lattice mismatch leads to strain energy in the crystals which must be relaxed by arrays of interfacial line defects in order for the interface to be stable. Finally, in the case of both grain and phase boundaries, there is accumulating evidence that the primary mechanism of boundary migration is the lateral motion of disconnections [7,8]. Disconnections are like interfacial dislocations embedded inside of an interfacial step of height h [9]; upon migration of the disconnection across the interface, the entire interface translates by a distance h.

While it has been well understood for some time that interfacial line defects play important roles in behaviors of materials, a comprehensive understanding of their properties is lacking. Most theories for interfacial line defects are based on continuum theories [1,9] which lack an accurate description of defect cores. Furthermore, these theories are unable to accurately characterize phenomena such as dislocation transmission and emission at interfaces because these processes are governed by localized nonlinear interatomic interactions [10]. On the other hand, molecular dynamics (MD) simulations have provided great insights into the core-dominated aspects of interfacial line defects, such as line defect reactions in interfaces [11,12], motion of interfaces containing line defects [13,14], and segregation of solutes to interfaces [15,16] One major challenge for MD in the context of studying interfaces is identifying and extracting interfacial line defects. Without the ability to readily identify interfacial line defects, quantify their Burgers vector content and step heights, and extract their positions within the interface, it is difficult or impossible to unambiguously analyze MD simulations of interfaces.

To date, there have been two classes of tools which have been developed for analysis of interfacial line defects in atomistic datasets: the dislocation extraction algorithm (DXA) which uses Burgers circuits to localize line defects and methods based on computation of the Nye tensor [17,18]. Compared to Nye tensor approaches, DXA has the

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benefit of unambiguously identifying the position and Burgers vector for each dislocation. DXA was originally developed to extract lattice dislocations and has seen widespread use since its invention [19,20]. DXA was subsequently extended to the case of interfacial dislocations [21], but has seen less use for this purpose. One reason for this lack of widespread adoption of DXA for interfacial analysis is that the user must fully specify the relaxed atomic structure within the interface. This relaxed structure is usually not known by the user, and often requires considerable effort to determine. On the other hand, Nye-tensor-based approaches do not require specification of this relaxed structure. Recently, Winters et al. applied the Nye tensor methodology to grain boundary dislocations and disconnections [22]. While powerful, this approach has the drawback that the user must specify the displacement shift complete lattice for the interface, and the method does not unambiguously identify the Burgers vector or line position. Finally, neither approach is able to detect and quantify the step height of a disconnection.

Our goal here is to develop a methodology for identifying and extracting interfacial line defects which is able to explicitly provide Burgers vectors and positions in space like DXA, but without requiring a significant amount of input from the user. Below we present a new approach to this problem that we call Interfacial Line Defect Analysis (ILDA), which uses Burgers circuits to analyze line defects like DXA but requires little or no input from the user on the nature of the interface. Additionally, ILDA enables automatic quantification of the step height h for disconnections. Below we present the ILDA algorithm and then demonstrate its performance on a pair of test cases: a $\Sigma 11$ twist GB and a phase boundary.

2. Theory

The ILDA algorithm differs from DXA in two key ways: (1) co-incidence sites are used to construct Burgers circuits which cross the interface and (2) a surface mesh is used to comprehensively explore interfaces for line defects and to insert defect line segments. The result is a tool which can identify interfacial line defects including disconnections without the need to specify the atomic structure within the interface.

Below we enumerate all of the steps in ILDA, which can be broken into four stages, as shown in Fig. 1. The algorithm is implemented as an open-source Python function using the OVITO Python API [23], and made available for interactive use in the OVITO Pro application [24] and integration in standalone Python codes/workflows [25]. All simulation graphics below were produced using OVITO Pro.

2.1. Stage 1: Identification of co-incidence sites

One of the challenges in identifying line defects in interfaces is dealing with the disordered atomic structure within the interface when performing Burgers circuits. DXA overcomes this challenge by requiring that the user specify the "perfect" atomic structure of the reference interface (without line defects) so that each circuit step within the interface can be mapped to a "perfect" step within the specified interface structure. Accordingly, Burgers circuits can weave arbitrarily through interfaces since the entire interface structure is known. Here, we adopt a simpler strategy: we first identify atoms that are located in or near co-incidence sites (CISs) between the two crystals. By definition, a CIS is a lattice site that is common to both crystals. This means that a Burgers circuit is able to cross the interface at a CIS without needing to know anything other than the structure of the crystals on each side. In ILDA, this is accomplished by performing half circuits through each crystal which connect at a pair of CISs. The CISs are thus "glue" which connects half circuits within the two crystals. Unlike DXA, only Burgers circuits which pass through co-incidence sites can be constructed using this method. The advantage is that CISs can be found in an automated manner, without requiring any knowledge about the interface structure. Furthermore, only CISs *within the interface* are required for the method to work. For example, the entire, three-dimensional co-incidence site lattice (CSL) does not need to be identified. In the case studies presented below, we employ the following algorithm to identify CISs:

Step 1. *Identify atoms of unknown structure.* With the help of the Polyhedral Template Matching (PTM) algorithm [26], identify atoms with an unknown ("Other") structure type. Most—if not all—atoms within the interface will have an unknown structure type, because PTM is designed to recognize only perfectly-coordinated lattice sites. PTM identifies structure types by comparing the polyhedron formed by an atom's local atomic neighborhood to a reference template for each crystal structure of interest. If the deviation from the closest matching reference template falls below a user-specified maximum root mean square deviation (RMSD), then it is declared to belong to that structure type. Atoms whose RMSD exceeds the maximum value for all structure types are declared to have unknown structure. Selection of the maximum RMSD value is thus an important aspect of using PTM and finding co-incidence sites.

Step 2. *Assign atoms to crystals.* Using the Grain Segmentation modifier in OVITO, assign all atoms to a crystal. The segmentation algorithm finds contiguous groups of atoms having the same crystalline structure type and a similar local lattice orientation, using information previously computed by the PTM algorithm. In the case of a single phase polycrystal, each crystallite is a grain, with grain boundaries separating crystals. The Grain Segmentation modifier works equally well for multiphase systems.

Step 3. *Find CIS atoms.* Out of the atoms with unknown structure type, identify atoms which are *simultaneously* polyhedral neighbors to two different crystals, i.e., an unknown atom has neighbor atoms in both crystals. Here "polyhedral neighbor atom" means an atom used to construct the polyhedron analyzed by PTM. The resulting set of atoms are by definition CIS atoms.

2.2. Stage 2: Construction of Burgers circuits

The heart of ILDA is the construction of Burgers circuits using three different coordinate frames, as shown in Fig. 2. First, a closed pair of half circuits are constructed in the Physical Frame. Each step within these circuits is mapped to the Lattice Reference Frames (one for each crystal) corresponding to the perfect crystal structures in the unit cell coordinate systems. This mapping is established by PTM's capability to associate neighbor atoms with corresponding lattice directions. Finally, the circuits are mapped from the Lattice Reference Frames to the Coherent Reference Frame (CRF) by rotating and/or stretching the crystals. The Coherent Reference Frame is the frame within which coherency is established between the two crystals. In the case of a grain boundary, the CRF enables definition of both the CSL and the displacement shift complete lattice [9]. In the case of a phase boundary, the CRF enables definition of a coherent dichromatic pattern used in the topological model for interfacial line defects [27,28].

Specification of the CRF is a tremendously important step, because it is within this frame that Burgers vectors are defined. The CRF can be defined by selecting a pair of deformation gradients, ${}_{\mathrm{A}}\mathbf{F}$ and ${}_{\mathrm{B}}\mathbf{F}$, which accomplish the mapping from the Lattice Reference Frames to the CRF. In ILDA, two options exist for defining ${}_{\mathrm{A}}\mathbf{F}$ and ${}_{\mathrm{B}}\mathbf{F}$. In cases where the user knows what CRF they want to use, it can be manually specified by inputting the crystal orientations and coherency strains. Consider the case where crystal A is oriented with crystallographic directions $\boldsymbol{x}_{\mathrm{A}}$, $\boldsymbol{y}_{\mathrm{A}}$, and $\boldsymbol{z}_{\mathrm{A}}$ pointed along the x, y, and z axes in the CRF, respectively.

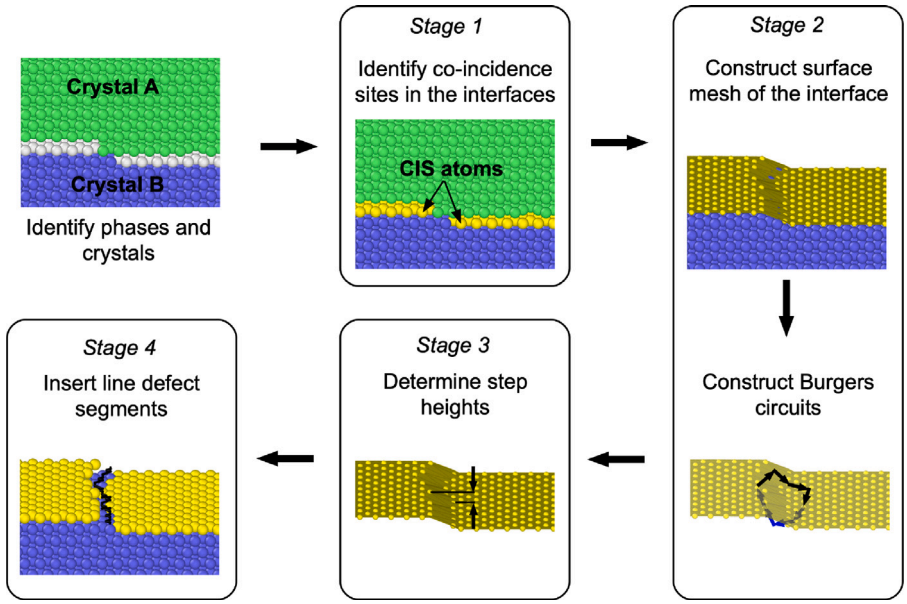


Fig. 1. Workflow for Interfacial Line Defect Analysis (ILDA), showing the four stages of the method. (CIS = co-incidence site).

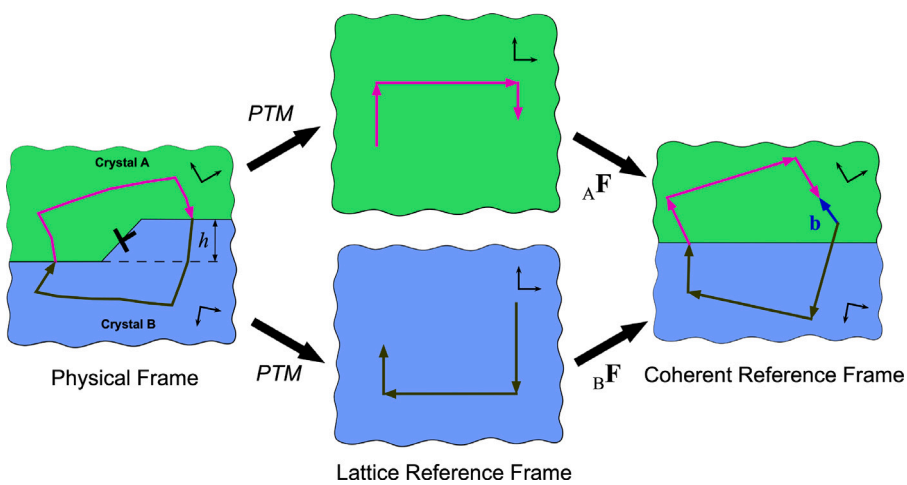


Fig. 2. Reference frames used by ILDA. In the Physical (simulation) Frame, closed circuits are constructed via a pair of half circuits (one in each crystal). Polyhedral template matching (PTM) then maps these to the Lattice Reference Frame for each crystal. A set of deformation gradients ${}_{\mathrm{A}}\mathbf{F}$ and ${}_{\mathrm{B}}\mathbf{F}$ then map to the Coherent Reference Frame where the Burgers vector is defined. The step height h is determined in the Physical Frame.

Also, the state of strain applied to establish coherency after rotating the crystal is ${}_{\mathrm{A}}\boldsymbol{\varepsilon}_c$. The result deformation gradient which maps to the CRF from the Lattice Reference Frame is

$$ {}_{\mathrm{A}}\mathbf{F} = (\mathbf{I} + {}_{\mathrm{A}}\boldsymbol{\varepsilon}_c) \cdot {}_{\mathrm{A}}\mathbf{R} \tag{1} $$

where ${}_{\mathrm{A}}\mathbf{R} = [\boldsymbol{x}_{\mathrm{A}}\ \boldsymbol{y}_{\mathrm{A}}\ \boldsymbol{z}_{\mathrm{A}}]$. When the user manually specifies the CRF, ILDA provides a robust means for identifying exact Burgers vectors, even if the physical interface in the simulation experiences a spatially varying strain state or set of line defects. If the user does not know a CRF, or if a quick estimate of the Burgers vectors is desired, ILDA can attempt to obtain estimates ${}_{\mathrm{A}}\mathbf{F}$ and ${}_{\mathrm{B}}\mathbf{F}$ by computing the local deformation gradients of the atoms near the interface. Specifically, a user-defined parameter d_{F} defines the thickness of the layer of atoms near the interface to be used. The atomic deformation gradients are computed using the method of Shimizu et al. [29], and then ${}_{\mathrm{A}}\mathbf{F}$ and ${}_{\mathrm{B}}\mathbf{F}$ are obtained as the average over all such atoms in each respective crystal. In our experimentation with ILDA, we find that these estimated values of ${}_{\mathrm{A}}\mathbf{F}$ and ${}_{\mathrm{B}}\mathbf{F}$ often yield reasonable estimates of the Burgers vectors, with errors of 10% or less. However, it should be noted that since ${}_{\mathrm{A}}\mathbf{F}$ and ${}_{\mathrm{B}}\mathbf{F}$ are approximate, the CRF is likely to be incoherent rather than coherent. This gives rise to spurious Burgers vectors that must be filtered out. Thus, another user-specified parameter b_{tol} dictates the minimum magnitude that a Burgers vector must have, otherwise it is ignored.

With the reference frames fully defined, it is possible to construct Burgers circuits in search of closure failures. However, it is unclear how these circuits can be performed in a comprehensive manner to ensure that all interfacial line defects are successfully identified. To accomplish this, we employ a surface mesh representation of the interface. Specifically, we construct a triangulated surface mesh whose vertices are CIS atoms in the interface. This surface mesh spans the whole interface, and provides a set of topological connections between the CIS atoms: the edges of the mesh. Thus, by constructing Burgers circuits between every pair of CIS atoms connected by an edge of the mesh, we can comprehensively explore the interface for line defects.

The following sequence of steps accomplishes the above.

Step 1. *Construct surface mesh.* A surface mesh which connects only CIS atoms is constructed using the so-called alpha-shape method in OVITO [30,31]. If any surface mesh elements

connect to non-CIS atoms, these elements are systematically "pruned" from the surface mesh by removing non-CIS atoms from the mesh.

Step 2. *Construct closed circuits*. For each edge of the surface mesh, construct a pair of half circuits which start and finish at the CIS atoms bounding the edge. Each successive atomic step in the half circuits is identified by choosing the atomic neighbor which minimizes the distance to the destination CIS atom. Let the vector connecting two atoms in the circuit in the Physical Frame be $\Delta \boldsymbol{x}_i$. Since the circuit is closed in the Physical Frame, $\sum_i \Delta \boldsymbol{x}_i = \boldsymbol{0}$. Each vector in the Physical Frame is mapped by PTM to a perfect lattice vector $\Delta \boldsymbol{X}_i$ in the Lattice Reference Frame.

Step 3. *Map lattice steps to coherent reference frame.* Each step in the Burgers circuit is mapped from the Lattice Reference Frame by computing the product ${}_{\mathrm{A}}\mathbf{F}\Delta \boldsymbol{X}_i$ for steps in crystal A and ${}_{\mathrm{B}}\mathbf{F}\Delta \boldsymbol{X}_i$ for steps in crystal B.

Step 4. *Determine Burgers vectors.* To determine the Burgers vector for each pair of half circuits, we simply sum each of the circuit step vectors in the CRF and compute the closure failure. Mathematically this is obtained as

$$\boldsymbol{b} = \sum_{i \in A} {}_{\mathrm{A}}\mathbf{F}\Delta \boldsymbol{X}_i + \sum_{j \in B} {}_{\mathrm{B}}\mathbf{F}\Delta \boldsymbol{X}_j \tag{2}$$

where the sums are over all circuit steps in the respective crystals.

2.3. Stage 3: Determination of step height

To determine step heights in the Physical Frame, we use the interface plane normal vector $\boldsymbol{n}$ associated with each Burgers circuit, as shown in Fig. 3.

Step 1. *Loop over mesh faces.* For an edge bounded by vertices i and j that has a non-zero Burgers vector (magnitude exceeding b_{tol}), loop over all mesh faces connected to vertices i and j.

Step 2. *Compute area vectors for defect-free faces.* For each face m connected to vertices i or j which does not contain any line defects (all edges of the face have Burgers vectors with magnitude less than b_{tol}), compute the area vector as

$$\boldsymbol{A}_m = \boldsymbol{v}_{ik} \times \boldsymbol{v}_{il} \tag{3}$$

where $\boldsymbol{v}_{ik}$ is the vector which points from vertex i to vertex k and face m is formed by vertices i, k, and l. Sum these area vectors as $\boldsymbol{A} = \sum_m \boldsymbol{A}_m$. We only use defect-free faces—shown as faces that are darkly colored in Fig. 3—for this computation because these faces are assumed to be associated with the "terrace plane" of the interface [28], within which coherency is established. In principle, this computation could be extended to include faces connected to neighbors of vertices i and j as well in order to get a more representative sampling of the interface.

Step 3. *Compute average normal vector.* Compute the average normal vector using all area vectors computed in Step 2 as

$$\boldsymbol{n} = \frac{\boldsymbol{A}}{\|\boldsymbol{A}\|} \tag{4}$$

Step 4. *Determine step height.* The step height is straightforwardly determined for each edge as

$$h = |\boldsymbol{n} \cdot \boldsymbol{v}_{ij}|. \tag{5}$$

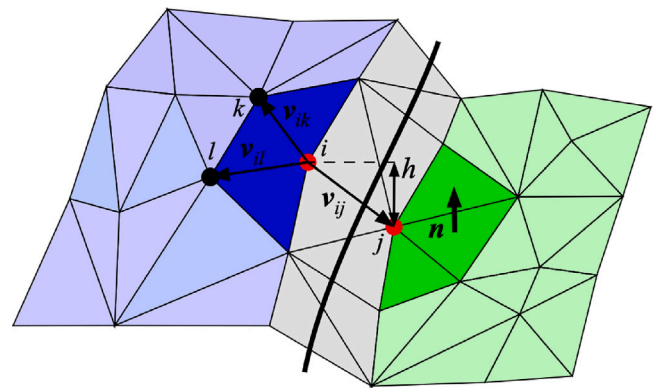


Fig. 3. Use of the interface surface mesh to determine disconnection step height at the edge connecting nodes i and j. Gray triangles are associated with the disconnection (heavy black line). Blue triangles are associated with the upper terrace and green triangles with the lower terrace of the interface, all of which are free of line defects. Darkly colored triangles are used to determine the habit plane normal vector $\boldsymbol{n}$ and the step height h using Eqs. (4) and (5). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2.4. Stage 4: Insertion of line segments

The aim of this last step is to construct a discrete line representation of the interfacial defects identified in the previous stages. In order to systematically and comprehensively insert line segments, we again utilize the surface mesh of the interface.

Step 1. *Check conservation of Burgers vector on mesh faces.* We loop over each face of the surface mesh and first check that Burgers vector conservation is satisfied. Each triangular mesh face can be thought of as a control volume which dislocation lines may "flow into" at its edges. The Burgers vector assigned to each edge gives the net Burgers vector content flowing (crossing) into that edge. To check Burgers vector conservation, we simply sum all Burgers vectors, being careful to ensure that all line sense vectors point into the mesh face (an arbitrary convention). If the sum is not close to zero within a user specified tolerance b_{tol}, the mesh face is flagged for the user as being problematic.

Step 2. *Insert line segments.* Since we have already verified that mesh faces properly satisfy Burgers vector conservation, line segment insertion is straight-forward. Segments are inserted in one of two ways, as shown in Fig. 4. If all edges have non-zero Burgers vectors, then segments are inserted which connect from the center of each edge to the centroid of the triangle. If only two edges have non-zero Burgers vectors, a segment is inserted connecting the centers of these edges. Subsequent smoothing operations may be applied to remove artificial jagged geometries, as is done in DXA.

To summarize: ILDA uses CIS atoms to construct a surface mesh, which is then utilized to construct a comprehensive set of Burgers circuits spanning the interface. Based on a user-specified CRF or an estimated CRF, Burgers vectors can then be defined and line segments inserted by applying conservation of Burgers vector to each mesh face. The interface structure and/or displacement shift complete lattice does not need to be defined for any of these steps.

2.5. Line defect statistics

Unlike with analysis of bulk line defects where the menu of possible Burgers vectors is known in advance, the set of possible line defects

Table 1
Simulation and coherent reference frame (CRF) coordinate axes used for the bicrystalline test cases considered here. (GB = grain boundary, NW = Nishiyama–Wasserman).

Case	Crystal	Phase	$\boldsymbol{x}_{\text{sim}}$	$\boldsymbol{y}_{\text{sim}}$	$\boldsymbol{z}_{\text{sim}}$	$\boldsymbol{x}_{\text{CRF}}$	$\boldsymbol{y}_{\text{CRF}}$	$\boldsymbol{z}_{\text{CRF}}$
Σ11 GB	A	FCC	$[146\ \overline{98}\ \overline{148}]$	$[542\ \bar{2}\ 536]$	$[\bar{1}\ \bar{3}\ 1]$	$[3\ \bar{2}\ \bar{3}]$	$[1\ 0\ 1]$	$[\bar{1}\ \bar{3}\ 1]$
	B	FCC	$[\overline{148}\ 98\ 146]$	$[\overline{536}\ \bar{2}\ \overline{542}]$	$[\bar{1}\ \bar{3}\ 1]$	$[\bar{3}\ 2\ 3]$	$[\bar{1}\ 0\ \bar{1}]$	$[\bar{1}\ \bar{3}\ 1]$
NW	A	FCC	$[1\ \bar{1}\ 0]$	$[1\ 1\ \bar{2}]$	$[1\ 1\ 1]$	$[1\ \bar{1}\ 0]$	$[1\ 1\ \bar{2}]$	$[1\ 1\ 1]$
	B	BCC	$[1\ 0\ 0]$	$[0\ 1\ \bar{1}]$	$[0\ 1\ 1]$	$[1\ 0\ 0]$	$[0\ 1\ \bar{1}]$	$[0\ 1\ 1]$

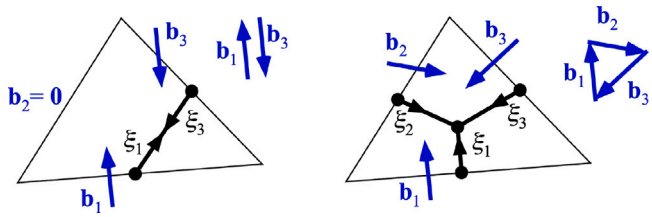


Fig. 4. Cases for line insertion using triangular mesh faces: (left) one segment is inserted because only two edges have non-zero Burgers vectors and (right) three segments are inserted because all three edges have non-zero Burgers vectors. In both cases, Burgers vector conservation is satisfied.

for a given interface is rarely known. This can make it difficult to assess the output of a line defect analysis code, such as ILDA. To help organize the output, ILDA tries to categorize the observed line defects. First, the Burgers vectors are sorted based on the magnitude of the Burgers vector; line defects with Burgers vectors of similar magnitude are grouped together. The b_{tol} parameter dictates how different two Burgers vector magnitudes have to be for them to be considered distinct from one another. Each distinct Burgers vector magnitude is assigned a "Burgers ID". Then, within each Burgers ID, the code sorts the Burgers vectors into "variants", with each variant corresponding to a Burgers vector with a distinct direction. For example, $[1\,1\,0]$, $[1\,0\,1]$, and $[\bar{1}\,1\,0]$ would be variants of the same Burgers ID. To determine whether two Burgers vectors are distinct, the angle between them is computed and if this angle is larger than a user-specified θ_{tol} parameter then they are declared separate variants. This analysis is especially useful when using an estimated CRF, because Burgers vectors are approximate and uncertain. In other words, Burgers vectors which should be identical are not. ILDA assigns Burgers IDs and variants to each Burgers vector, and then computes the average and standard deviation (weighted by the segment lengths) for each component of each Burgers vector within the same Burgers ID and variant. These values are outputted at the end of the analysis. Judicious choice of b_{tol} and θ_{tol} can lead to a reasonable understanding of the line defect content in an interface, even with the uncertainty introduced by an estimated CRF.

3. Results

We demonstrate ILDA's functionality on two different test cases: one GB and one biphase interface. Both cases consider a planar interface between two periodic crystals. The crystal orientations used to construct the atomistic systems are given in Table 1 (the 'sim' vectors). In this table, we also give the crystal orientations used to specify the CRF. Simulations were performed using LAMMPS [32]. The run-time for the Python implementation of ILDA in OVITO was on the order of 5 min on a laptop computer for these interfaces containing $\sim 10^4$ CIS atoms.

3.1. Σ11 Twist GB

A GB vicinal to the Σ11(131) 180° pure twist boundary with face-centered cubic (FCC) crystals was considered. Crystals were inserted in a non-orthogonal simulation cell resulting in a 179.29° twist about the (131) axis between the top and bottom crystals. The CRF for this interface corresponds to the grain orientations for the perfect Σ11 GB. Since this is a GB, no coherency strains need to be imposed to achieve coherency in the CRF. The system contained about 430,000 atoms and the interface was approximately 312 Å by 345 Å. After inserting the crystals, the system was relaxed via energy minimization, annealed at 400 K for 24 ps, and then relaxed again via energy minimization for line defect analysis. The Mishin et al. embedded atom method potential for Cu was used [33].

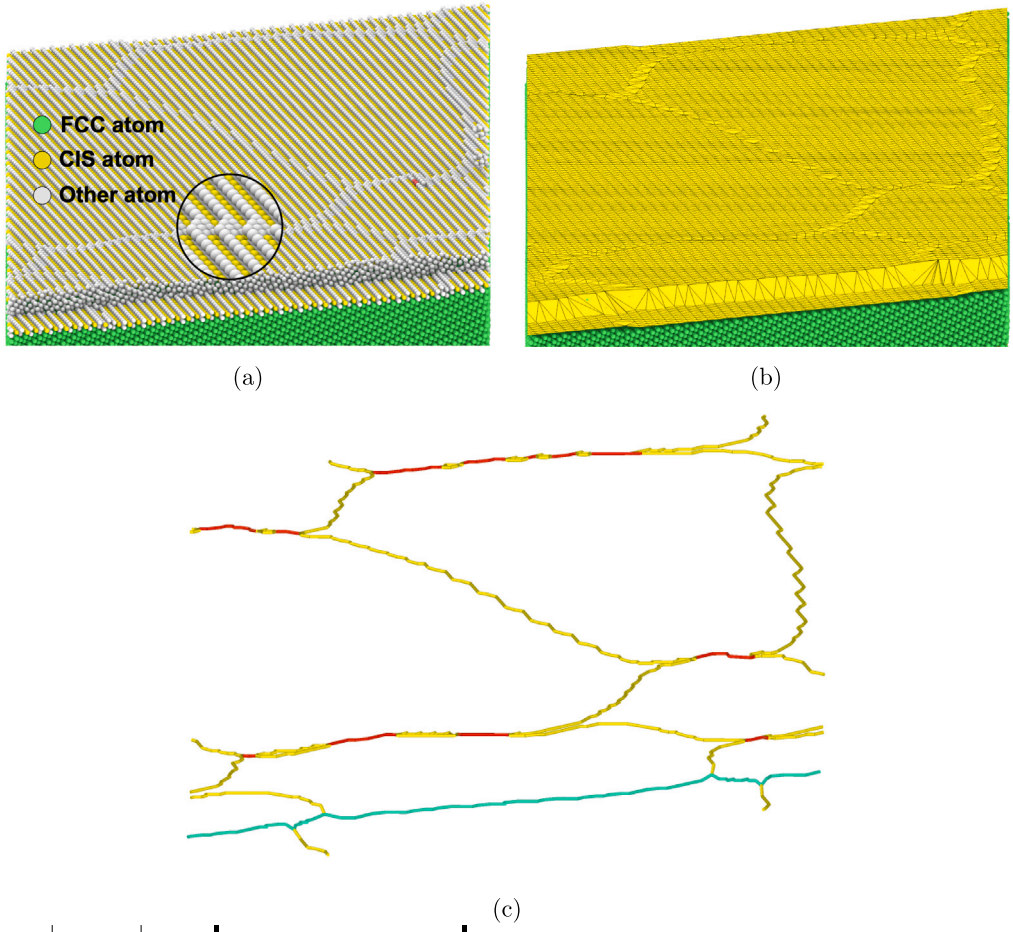
Fig. 5 shows the results. First, Fig. 5(a) shows that the atomic structure of the interface is not flat, but instead contains numerous curved steps, indicating the possible presence of disconnections. These defects nucleated to accommodate the misorientation from the perfect 180° twist. Using the CIS identification subroutine, 6011 CIS atoms were identified using a maximum RMSD value of 0.06 Å, and these atoms are colored yellow. These CIS atoms form an approximately planar interface in between the stepped defects. Note that only a fraction of the atoms in the GB were identified as CIS atoms (white atoms in the interface have unknown structure and are not CIS atoms). Fig. 5(b) shows the surface mesh resulting from these CIS atoms. Fig. 5(c) finally shows the line defect structure identified by ILDA using the user-specified CRF given in Table 1. Line defects are colored based on their Burgers ID, each Burgers ID corresponds to a distinct Burgers vector magnitude. The Burgers vectors and the average step height $\langle h \rangle$ for each are given in the table in Fig. 5, which also provides a color legend for the line defects. We have analyzed the same interface using DXA and find agreement between ILDA and DXA on both line configurations and Burgers vectors. Recall that while DXA requires the interfacial atomic structure to be specified, ILDA does not.

The majority of the line defect network is comprised of "ID 2" disconnections with a small step height of 1.09 Å (yellow lines). ID 2 disconnections have two variants, whose Burgers vectors only differ by a single negative sign. When one ID 2 disconnection from each variant meets, they react to form an ID 3 disconnection with a much smaller Burgers vector and larger step height (red lines). However, this reaction does not seem to be very stable, because we see many locations where the structure stays disassociated as two yellow ID 2 lines rather than combining together into one red ID 3 line. At the bottom of the figure, we also see ID 1 disconnections (green lines) with a large step height of 5.71 Å. These form when five or six ID 2 disconnections combine together, and three distinct variants are observed.

In the table in Fig. 5, we also list the average Burgers vectors and associated standard deviations obtained using the estimated CRF computed with $d_{\text{F}} = 5$ Å, $b_{\text{tol}} = 0.5$ Å, and $\theta_{\text{tol}} = 10°$. Approximately 18,700 atoms from each crystal were used to approximate ${}_{\text{A}}\mathbf{F}$ and ${}_{\text{B}}\mathbf{F}$. We can see that the Burgers vectors obtained using the estimated CRF are generally accurate to within 0.1 Å and the standard deviations are all less than 0.04 Å. These estimated Burgers vectors were obtained with no user input on the geometry of the problem or the nature of the interface.

3.2. Nishiyama–Wasserman interface

Next we consider a biphase interface using the Zhou et al. embedded atom method potential for Fe [34], examining an FCC crystal coherently bonded to a body-centered cubic (BCC) crystal. The two crystals satisfy the Nishiyama–Wasserman (NW) orientation relation, with $(111)_{\text{FCC}} \parallel (110)_{\text{BCC}}$ for the interface plane normal, and $[\bar{1}10]_{\text{FCC}} \parallel [\bar{1}00]_{\text{BCC}}$ and $[11\bar{2}]_{\text{FCC}} \parallel [01\bar{1}]_{\text{BCC}}$ in the plane of the interface; this set of



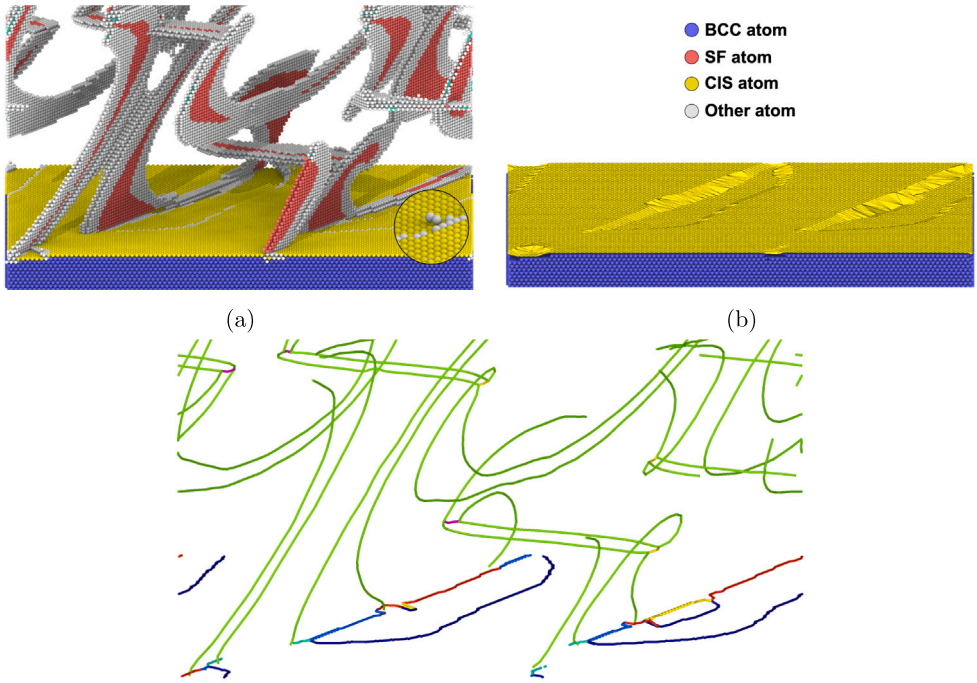
			Exact CRF			Estimated CRF		
ID	Color	$\langle h \rangle$	b_x	b_y	b_z	b_x	b_y	b_z
1		5.24	2.312	0	0	2.330 ± 0.007	-0.027 ± 0.034	0.004 ± 0.005
		6.63	1.927	1.278	0	1.965 ± 0.014	1.188 ± 0.032	0.012 ± 0.004
		6.58	1.927	−1.278	0	1.956 ± 0.015	-1.220 ± 0.038	-0.008 ± 0.005
2		1.09	0.385	1.278	0	0.342 ± 0.020	1.219 ± 0.040	0.008 ± 0.005
		1.09	0.385	−1.278	0	0.325 ± 0.017	-1.254 ± 0.049	-0.002 ± 0.007
3		2.24	0.771	0	0	0.671 ± 0.007	-0.015 ± 0.041	0.001 ± 0.005

Fig. 5. Line defect analysis for a GB vicinal to a $\Sigma 11$ twist GB. (a) Atomic view with FCC atoms from the top grain removed. Yellow atoms are CIS atoms. Inset shows zoom of the interface. (b) Surface mesh used by ILDA. (c) Line defects extracted by ILDA. The table gives information on the 3 distinct interfacial line defects and their variants using exact and estimated coherent reference frames (CRFs). (CIS = co-incidence site). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

crystallographic relationships also defines the CRF for the interface. The NW orientation relation has been observed experimentally for martensitic interfaces in steels [35]. Since this is a biphase interface, coherency strains are necessary to achieve coherency in the CRF; specifically, for lattice parameters $a_{\mathrm{FCC}} = 3.58$ Å, $a_{\mathrm{BCC}} = 2.87$ Å and an interface plane orthogonal to the z-axis, the following set of coherency strains are specified:

$$ {}_{\mathrm{A}}\boldsymbol{\epsilon}_c = \begin{bmatrix} 0.0669 & 0 & 0 \\ 0 & -0.0372 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad {}_{\mathrm{B}}\boldsymbol{\epsilon}_c = \begin{bmatrix} -0.0590 & 0 & 0 \\ 0 & 0.0401 & 0 \\ 0 & 0 & 0 \end{bmatrix} $$

where crystal A is the FCC crystal and crystal B is the BCC crystal. This set of coherency strains results in a CRF that is halfway between the unstrained FCC and BCC crystals (e.g., the lattice spacings in the interface are the average of the FCC and BCC spacings), as used by Ma and Pond [28]. Note that while this choice is arbitrary, a "correct" CRF does exist such that the long-range stress fields of the interface are zero when the line defects at the interface satisfy the Frank–Bilby equation [36]. Determination of the "correct" CRF requires solution of an elasticity problem [36], which is beyond the scope of this paper. However, once the Burgers vectors are determined in *any* CRF, they can be easily mapped to a different CRF by simply applying a deformation gradient. Consider the following example. Let $\boldsymbol{b}$ be a Burgers vector obtained in the exact CRF given above, where coherency strains are equally partitioned between crystals A and B. According to Eq. (2), this Burgers vector can be expressed in a new CRF where all coherency strain has been applied to crystal B as $\boldsymbol{b}' = (\mathbf{I} + {}_{\mathrm{A}}\boldsymbol{\epsilon}_c)^{-1} \cdot \boldsymbol{b}$, which effectively "removes" the coherency strain from crystal A and instead applies it to crystal B.



			Exact CRF			Estimated CRF		
ID	Color	$\langle h \rangle$	b_x	b_y	b_z	b_x	b_y	b_z
1		2.01	1.351	1.407	0.038	1.271 ± 0.017	1.388 ± 0.030	0.050 ± 0.004
2		3.99	1.351	0.704	0.075	1.275 ± 0.006	0.816 ± 0.010	0.081 ± 0.001
3		1.85	0.901	0.235	−0.651	0.853 ± 0.011	0.246 ± 0.018	-0.649 ± 0.002
4		1.97	0	0.704	−0.038	-0.024 ± 0.010	0.537 ± 0.018	-0.031 ± 0.002

Fig. 6. Line defect analysis for Shockley–Read partial dislocations impinging on an FCC-BCC NW interface. (a) Atomic view with FCC atoms removed. Yellow atoms are CIS atoms. Inset shows zoom of the interface. (b) Surface mesh used by ILDA. (c) Line defects extracted by ILDA and DXA. The table gives information on the 4 distinct interfacial line defects identified by ILDA. (SF = stacking fault, CIS = co-incidence site, NW = Nishiyama–Wasserman). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The biphase system contains about 3 million atoms and is constructed to form a 220 Å by 270 Å interface, which is relaxed with a zero-pressure NPT ensemble at 5 K. A set of three glide dislocation loops of dimensions 60 Å by 100 Å are also introduced into the FCC crystal by carefully removing parallelepipedal regions of atoms. During relaxation, these dislocation loops bowout and impinge on the interface, stimulating the nucleation of stepped defects in the interface, as shown in Fig. 6(a). In this figure, the 12,730 identified CIS atoms, as obtained using a maximum RMSD value of 0.05 Å, are shown as yellow. All atoms in the interface are CIS atoms for this test case. Fig. 6(b) shows the surface mesh which spans these CIS atoms. In order to fully assess the line defects in this simulation, we utilize DXA to analyze the bulk dislocations and ILDA to analyze the interfacial line defects. Fig. 6(c) shows the line defect structure identified by DXA and ILDA using the user-specified CRF given in Table 1. Four distinct interfacial line defects are identified, which are listed in the table in Fig. 6, each only has one variant. This analysis clearly reveals that the impingement on the interface by a Shockley–Read partial dislocation (colored green) leads to disassociation into a pair of disconnections, one with ID 3 (colored light blue) and one with ID 4 (colored dark blue). The ID 4 disconnection appears to be glissile and begins gliding away from the original intersection area, whereas the ID 3 disconnection appear to be immobile. After a second Shockley–Read dislocation hits the same region on the interface, a Type 1 disconnection (red line) is formed. Type 1, 3, and 4 disconnections all have step heights around 2 Å. Looking to the bottom right of the figure, we see that the Type 1 disconnection is able to emit another Type 4 disconnection, forming a new Type 2 (yellow) disconnection with a much larger step height of about 4 Å.

The Burgers vector statistics obtained using the estimated CRF with $d_F = 5$ Å, $b_{tol} = 0.25$ Å, and $\theta_{tol} = 10°$ are also given in the table in Fig. 6. Approximately 26,000 atoms were used from each crystal to compute the estimated values of ${}_A\mathbf{F}$ and ${}_B\mathbf{F}$. When it comes to estimating the CRF from the interface in the simulation, this test case is more challenging since the FCC crystal contains numerous line defects which are impinging on the interface, whereas no such defects were present in the $\Sigma 11$ GB case. Compared to the twist GB example, these estimated Burgers vectors differ more significantly from their exact CRF counterparts. These differences are a result of the different rotation matrices and coherency strain tensors which define the estimated CRF. To better understand this, we employ polar decomposition to decompose the estimated deformation gradients into their corresponding rotations $\mathbf{R}$ and stretches $\mathbf{V} = \mathbf{I} + \boldsymbol{\varepsilon}_c$, such that $\mathbf{F} = \mathbf{V} \cdot \mathbf{R}$ [37]. The estimated deformation gradients for this test case are:

$$ {}_A\tilde{\mathbf{F}} = \begin{bmatrix} 0.075 & 0.006 & -0.753 \\ 0.410 & 0.764 & 0.423 \\ 0.580 & -0.580 & 0.577 \end{bmatrix}, \quad {}_B\tilde{\mathbf{F}} = \begin{bmatrix} 0.943 & -0.001 & 0 \\ 0.001 & 0.741 & 0.723 \\ 0 & -0.711 & 0.711 \end{bmatrix}. $$

The polar decomposition of these deformation gradients gives the following estimated coherency strain tensors:

$$ {}_{\mathrm{A}}\tilde{\boldsymbol{\epsilon}}_c = \begin{bmatrix} 0.062 & -0.005 & 0 \\ -0.005 & -0.035 & 0.020 \\ 0 & 0.020 & 0.003 \end{bmatrix}, \quad {}_{\mathrm{B}}\tilde{\boldsymbol{\epsilon}}_c = \begin{bmatrix} -0.057 & 0.001 & 0 \\ 0.001 & 0.036 & 0.006 \\ 0 & 0.006 & 0.006 \end{bmatrix}. $$

Comparing these estimated coherency strains with the exact ones above, we see several differences. First, several spurious shear strain components with magnitudes on the order of 0.01 appear in the estimated strain tensors. And second, the coherency strain magnitudes in the estimated tensors are uniformly smaller than those of the exact tensors (e.g., 0.062 compared with 0.0669). Both of these effects give rise to a slightly incoherent interface in the estimated CRF, and cause the Burgers vectors to differ from the exact CRF.

4. Discussion

The major benefit of ILDA compared to DXA is that little, or no, information about the nature of the interface is necessary to use it. In the case where a user knows the CRF for the interface—for example when a manually constructed planar interface is being studied (as in our test cases)—then the user simply needs to specify the known crystal orientations and coherency strains that define the CRF. The line defects that are extracted in this case are precisely defined with respect to that CRF. This scenario applies to many previous works, such as the large body of research focused on GB mobility, energy, and GB interactions with bulk line defects [38], and the large but more modest body of work on planar biphase interfaces. In the case where a user does not know the CRF, the estimated CRF allows an initial analysis without having to specify *anything* about the interface. Of course the results using an estimated CRF are only approximate, and some fine-tuning may be required to determine the appropriate values of d_{F}, b_{tol}, and θ_{tol} which provide useful results. Furthermore, the user may be able to use the estimated CRF to help determine the true CRF using the polar decomposition, as shown in Section 3.2. ILDA automatically performs this polar decomposition for the user and reports the results. We believe that the ability to approximately identify interfacial defects without any user input is a major advancement for the MD community.

There is, of course, a strong requirement before ILDA can be used: it must be possible to locate CIS atoms within the interface. If CIS atoms cannot be located, the method is unable to find line defects. In the cases presented above, there was a dense set of CIS atoms which made it possible to extract line defects with high spatial resolution (close to the atomic spacing). However, if the interface is more heavily disordered, ILDA may fail to identify CIS atoms. This may be because: (1) CIS atoms do not exist for the interface (e.g., a general GB), (2) the interface is too disordered to identify CIS atoms, or (3) CISs exist but atoms do not reside at them. In case (1), there is no way to define interfacial defects anyway, so the problem is moot. In case (2), the disorder may be due to elevated temperature, in which case quenching the system or minimizing the energy may reveal the CIS atoms. Or, the disorder may simply be a property of the interface. For example, some GBs may have a structure which is several atomic planes thick. CIS atoms may be difficult to identify in such interfaces. In case (3), the ILDA algorithm would need to be extended in order to account for the possibility of "virtual" CISs, e.g., CISs where no atoms reside; this is possible in principle, but additional research is necessary. Beyond these scenarios, there may be cases where CIS atoms can be located, but they are spaced so far apart from each other that it is not possible to locate line defects with high spatial resolution. Of course the defect content identified by ILDA will still be valid, it is just that each line segment inserted by ILDA may be a composite from many different line defects that exist in between the CIS atoms. Finally, we argue that identification of CISs is a valuable capability by itself, since so much of interfacial structure and properties are related to the co-incidence site lattice of an interface. The ability to automatically identify and analyze these sites for arbitrary bicrystals could be valuable to the study of interfaces beyond the search for line defects.

It is worth reiterating the differences between ILDA and DXA. Both approaches use Burgers circuits to identify line defects. The major differences are: (1) how interfaces are handled during Burgers circuit construction and (2) how line segments are inserted.

Regarding (1), in DXA interfaces are treated exactly the same as bulk crystalline regions. A "perfect" atomic structure is identified for bulk and interface regions via a repeating atomic motif. Accordingly, any atomic jump inside of an interface could be used in constructing Burgers circuits. In contrast, ILDA treats interfaces fundamentally differently than bulk regions. Only the CIS atoms within an interface are used to construct Burgers circuits, other interface atoms are ignored. As a result, ILDA will not be able to locate line positions within the interface as precisely as DXA, but also does not require the specification of a structural motif for the interface.

Regarding (2), in DXA line segments are inserted by sweeping a single Burgers circuit which is known to have a closure failure along the manifold surrounding regions containing atoms of unknown structure type. In ILDA, line defects are inserted by applying conservation of Burgers vector to each face of the interfacial surface mesh individually (see Fig. 4). Unfortunately this difference in line insertion means that the line defects extracted by ILDA and DXA cannot presently be tied together at their intersections points; additional research is necessary to properly couple them together. Such a unified algorithm may be enabled by a future reformulation of DXA based on a 3D Delaunay tessellation in which short dislocation segments are constructed in each tetrahedral element [39]. Another point to make is that, in principle, the ILDA approach should be more amenable to parallelization than DXA, since all Burgers circuit construction and line insertion events are independent of each other. An additional benefit of the interfacial surface mesh approach of ILDA is the ability to extract the step height of a disconnection, which is not currently possible in DXA.

Interestingly, the notion of a CRF as used in ILDA is quite similar to the reference frames and transformation matrices used in DXA to construct Burgers circuits that span interfaces. The only substantive difference is that in ILDA mismatched interfaces with coherency strains are allowed. We note that DXA is quite more general than ILDA in terms of being able to construct Burgers circuits which span disparate atomic structures. For example, in DXA a Burgers circuit could be constructed around a GB triple point, which requires crossing three separate interfaces. As introduced above, ILDA cannot identify line defects at triple points. Perhaps ILDA and DXA could be further combined by coupling the transformation matrices concept from DXA with ILDA's use of CIS atoms and an interfacial surface mesh.

A final, and rather subtle difference between ILDA and DXA is in the nature of the reference state(s) used to define the Burgers vector. In DXA, the reference state used for Burgers circuit construction is obtained via energy minimization of the interfacial structure. In other words, the atomic jumps within the interface correspond to the atomic positions in the true, relaxed interface structure. On the other hand, ILDA only uses CISs within the interface which are defined in the unrelaxed state. This is akin to how a CSL is defined as a theoretical lattice based on abutting two perfect crystals together. The atoms in a real interface which derives from the co-incidence site lattice may displace away from their theoretical positions during relaxation.

5. Conclusions

We have presented a new methodology for identifying and extracting interfacial line defects, including dislocations and disconnections, which requires little or no user input on the nature of the interface. This method, which we have called ILDA, is similar to DXA in its performance, but unlike DXA relies on the ability to find CIS atoms in the interface. We apply the approach to two test cases: a pure twist GB vicinal to a Σ boundary and an FCC-BCC interface. In both cases

ILDA was able to reveal in detail what line defects were present in the interface, both in terms of their Burgers vectors and their step heights. Notably, by using the atoms near the interface to estimate the CRF, accurate estimates of the Burgers vectors could be obtained without specifying anything about the interface. In the future, we hope to improve the coupling between ILDA and DXA to lead to a unified methodology for line defect analysis which is easy to use and efficient for high-performance computing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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